LaTeX Documentation Examples

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Abstract

This document demonstrates the main features of LaTeX: structure, text formatting, lists, mathematics, tables, figures, references, page layout, custom commands and code listings.

Contents

1 Structure and Text **2**
1.1 Document structure . . . 2
1.1.1 A subsection . . . 2
1.2 Text formatting . . . 2

2 Lists and Mathematics **3**
2.1 Lists . . . 3
2.2 Mathematics . . . 3

3 Tables and Figures **5**
3.1 Tables . . . 5
3.2 Figures . . . 5
3.2.1 Subfigures . . . 5

4 References, Code and Commands **7**
4.1 Cross-references and citations . . . 7
4.2 Code listings . . . 7
4.3 Custom commands . . . 7

A Appendix Example **9**

Chapter 1

Structure and Text

1.1 Document structure

A LaTeX file has a *preamble* (before `\begin{document}`) and a *body*. Sectioning commands: `\chapter`, `\section`, `\subsection`, `\subsubsection`, `\paragraph`.

1.1.1 A subsection

Text of the subsection.

A subsubsection

Text of the subsubsection.

1.2 Text formatting

Bold, *italic*, underline, `typewriter`, Small Caps, *emphasis*.

tiny small normal large Large huge

Centered text

Right-aligned text

Special characters must be escaped: $ % & # _ { } .

A new paragraph starts after a blank line.
This line starts after a forced line break.
No indentation here. A horizontal gap.

Chapter 2

Lists and Mathematics

2.1 Lists

- First item
- Second item
  - Nested item

1. Step one
2. Step two

Git Version control system.

GitHub Hosting platform for Git repositories.

2.2 Mathematics

Inline math: $E = mc^2$ and $\alpha + \beta = \gamma$.

Display math with numbering:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{2.1}$$

Equation 2.1 is the quadratic formula.

Unnumbered display:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Aligned equations:

$$(a+b)^2 = a^2 + 2ab + b^2 \tag{2.2}$$
$$(a-b)^2 = a^2 - 2ab + b^2 \tag{2.3}$$

Integral and limit:

$$\int_0^\infty e^{-x}\,dx = 1, \qquad \lim_{x \to 0} \frac{\sin x}{x} = 1$$

Matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Piecewise function:

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

Chapter 3

Tables and Figures

3.1 Tables

Left	Center	Right
A	B	C
Long text	12	3.14

Table 3.1: A basic table

Professional style with `booktabs`, `multicolumn` and `multirow`: Tables 3.1 and 3.2

Tool		Year
Version control	Git	2005
	SVN	2000
Documentation	Doxygen	1997

Table 3.2: A table using booktabs

show two table styles.

3.2 Figures

LaTeX can draw graphics itself using TikZ. A figure with a caption: See Figure 3.1.

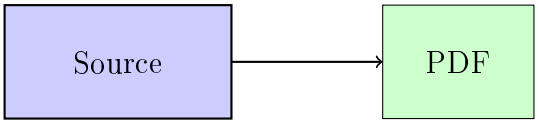


Figure 3.1: A simple TikZ diagram

To include an image file use `\includegraphics[width=0.6\textwidth]{image.png}` (needs `graphicx`).

3.2.1 Subfigures



(a) Circle (b) Square

Figure 3.2: Two subfigures

Chapter 4

References, Code and Commands

4.1 Cross-references and citations

Use `\label` and `\ref` for cross-references: Chapter 4.3 is on custom commands. Citations use BibTeX: the standard references are [2, 1], and information theory started with [3]. Hyperlinks: `https://www.latex-project.org/` or Overleaf. Footnote example.[1]

4.2 Code listings

Listing 4.1: A Python listing

```
1 def add(a, b):
2     return a + b
3
4 print(add(2, 3))
```

Inline code: `git commit -m "message"`.

4.3 Custom commands

Commands defined in the preamble: **Git**, **LaTeX**, and a vector $\mathbf{x}$. Syntax: `\newcommand{\name}[num a`

[1] A footnote appears at the bottom of the page.

Bibliography

[1] Donald E. Knuth. *The TEXbook*. Addison-Wesley, 1984.

[2] Leslie Lamport. *LATEX: A Document Preparation System*. Addison-Wesley, 2nd edition, 1994.

[3] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

Appendix A

Appendix Example

Appendices use `\appendix` and then `\chapter` or `\section`.